

Communication

Date	1 Jul 2026
Team ID	XXXXXX
Project Name	Smart Crop
Maximum Marks	1 Mark

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Team Standup	Daily	Slack / WhatsApp	Project Team Members	Sync daily tasks, discuss immediate goals, and ensure alignment on OptiCrop algorithm integrations (N, P, K parameters).
2	Progress Update	Weekly	Progress Update Weekly Zoom / MS	Project Team & Mentor	Review weekly developments on the software engine, evaluate data integration modules, and check milestone timelines.
3	Issue / Bug Discussion	As Needed	GitHub / Discord	Developers & QA Team	Track and resolve software bugs, data inconsistencies, or anomalies in crop recommendation outputs.
4	Stakeholder Review	Bi-Weekly	Google Meet	Team, Mentors & Stakeholders	Share progress on the optimization engine, present environment compatibility features, and collect strategic feedback.
5	Final Demo Rehearsal	Once	MS Teams	Entire Project Team	Conduct a dry-run of the 3 key scenarios (Smart Crop Recommendation, Suitability Assessment, and Policy Planning) before final submission.

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Data Complexity Alignment: Team members had conflicting approaches on how to weight and integrate complex environmental factors (N, P, K levels, temperature, soil pH, and rainfall) into a unified analysis matrix.	Resolution: Established a dedicated technical sync session to standardize the data preprocessing guidelines. We locked in fixed parameters for Scenario 1 and Scenario 2, ensuring seamless formula collaboration across the development team.
2	Tracking Research Requirements: Difficulty bridging the gap between practical farming metrics for everyday farmers (Scenario 1) and macro data-driven analytics required by researchers and policymakers (Scenario 3).	Resolution: Conducted a specialized scoping review session. The team separated the UI/UX pathways—creating a simplified dashboard for instant crop recommendations alongside an advanced analysis module for resource optimization strategies.
3	Remote Integration Obstacles: Minor communication gaps during asynchronous code merges on GitHub led to temporary integration errors within the predictive engine.	Resolution: Mandated the use of clear, structured pull-request templates and quick Discord audio huddles whenever a conflict occurred. This successfully streamlined code reviews and guaranteed a stable final platform build.